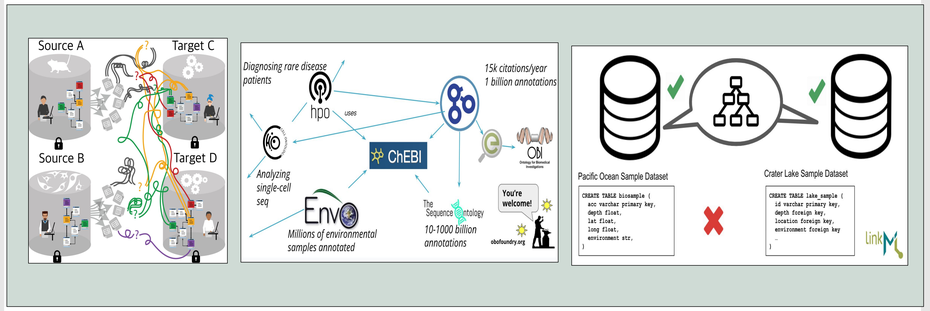
LinkML: The Linked Data Modeling Language Framework

Sierra Moxon[1], J. Harry Caufield[1], Corey Cox[2], Nomi Harris[1], Harshad Hegde[1], Patrick Kalita[1], Mark Miller[1], Sujay Patil[1], Tim Putman[2], Justin Reese[1], Kevin Schaper[2], Harold Solbrig[3], Christopher J. Mungall[1]

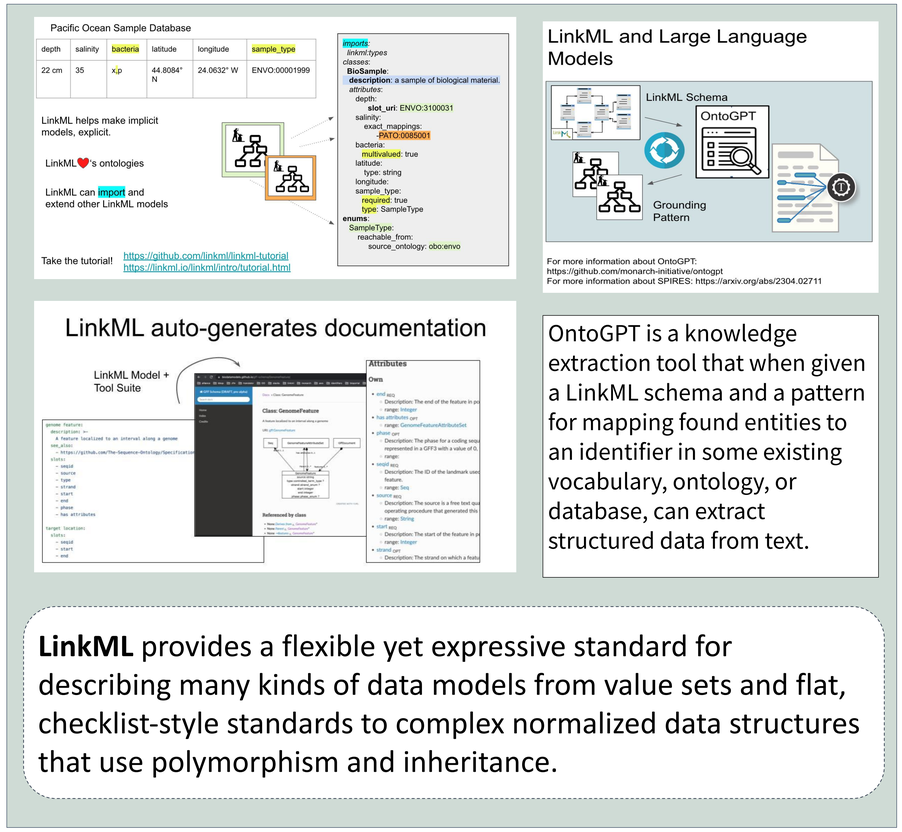
[1]Lawrence Berkeley National Laboratory, [2]University of Colorado Anschutz Medical Campus, [3] Johns Hopkins University

Why do we need LinkML?



In the Biomedical domain, data often comes in many shapes and sizes. Its dispersed between many different resources and much of the data is captured in semi-structured formats (free text, notes, figures, publications, lab notebooks, spreadsheets). Even with shared standards, from different knowledge bases requires schema wrangling.

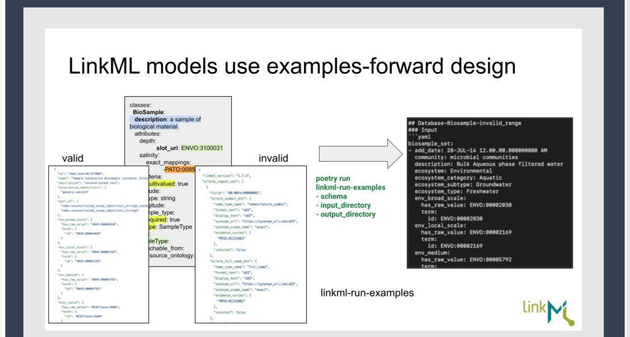
LinkML makes implicit models, explicit

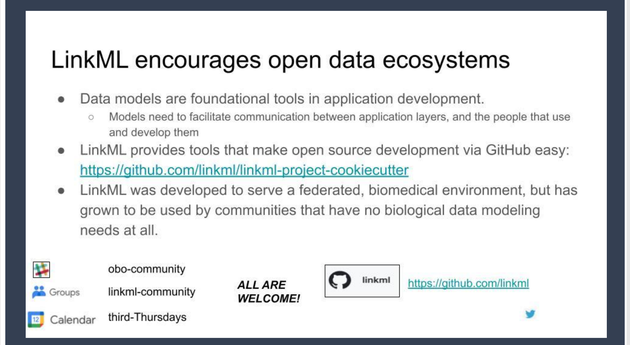


OntoGPT is a knowledge extraction tool that when given a LinkML schema and a pattern for mapping found entities to an identifier in some existing vocabulary, ontology, or database, can extract structured data from text.

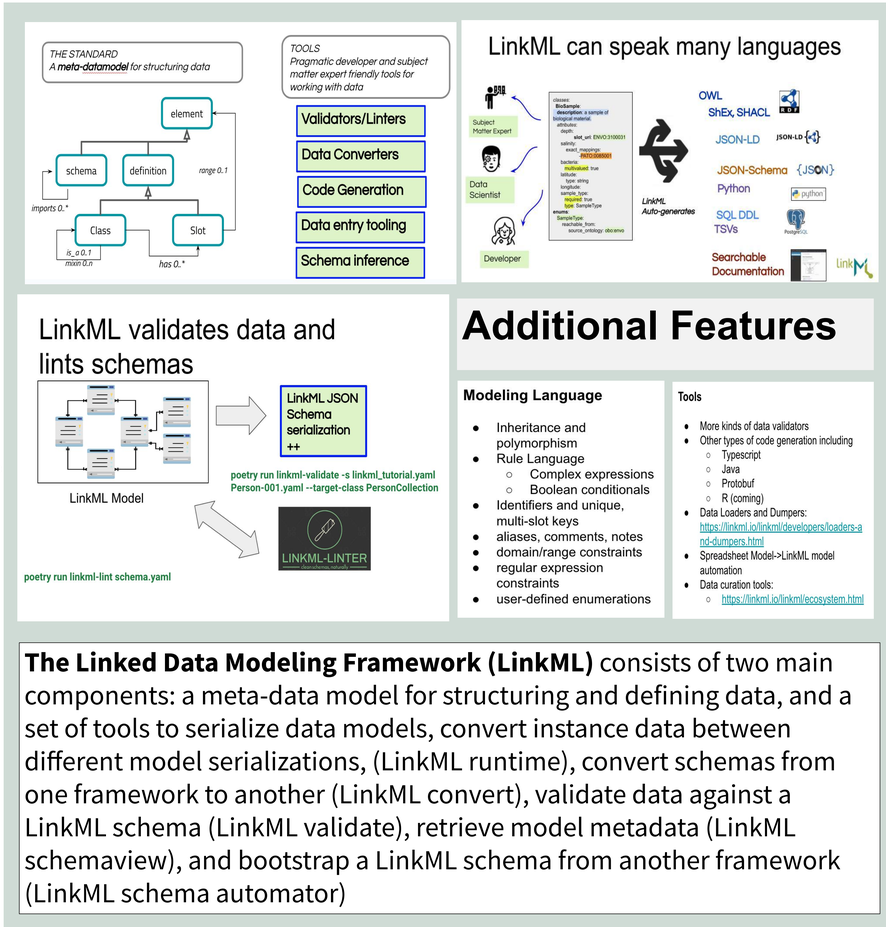
LinkML provides a flexible yet expressive standard for describing many kinds of data models from value sets and flat, checklist-style standards to complex normalized data structures that use polymorphism and inheritance.

Key insights from early LinkML adoption





LinkML has a simple but powerful YAML specification, and a robust toolchain



The Linked Data Modeling Framework (LinkML) consists of two main components: a meta-data model for structuring and defining data, and a set of tools to serialize data models, convert instance data between different model serializations, (LinkML runtime), convert schemas from one framework to another (LinkML convert), validate data against a LinkML schema (LinkML validate), retrieve model metadata (LinkML schemaview), and bootstrap a LinkML schema from another framework (LinkML schema automator)

Discussion

LinkML is a data modeling framework that provides a flexible yet expressive standard for describing many kinds of data models. It is purposefully designed so that software engineers and subject matter experts can communicate effectively in the same language while also providing the semantic underpinnings to make data conforming to LinkML schemas easier to understand and reuse computationally.

For more information about **LinkML**: **https://github.com/linkml/linkml**

Contact Us: linkml-discuss@lbl.gov

slack: obo-communitygroup.slack.com

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